Dynamic Inventory Risk & Reorder Engine: An End-to-End Operational Analytics Pipeline

Jacob C Sellinger

Northwest Missouri State University, Maryville MO 64468, USA
S577258@nwmissouri.edu

Abstract. WORK IN PROGRESS
This capstone project presents an end-to-end data engineering and analytics pipeline designed to optimize supply chain inventory under lead-time uncertainty. By processing raw transactional data using DuckDB, predicting supplier lead times with machine learning, and calculating dynamic safety stock levels, the engine surfaces real-time stockout risks via an interactive Python Streamlit dashboard.

Keywords: Supply Chain Analytics · Lead Time Prediction · Dynamic Reorder Point · DuckDB · Streamlit.

1 Introduction

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Modern supply chains face significant volatility in lead times and demand. Traditional static reorder point models often fail under stochastic conditions, resulting in costly stockouts or excessive holding costs...

2 Methodology Overview

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The system architecture consists of a three-stage pipeline: ingestion and cleaning (ETL), predictive lead time modeling, and interactive decision support...

2.1 Mathematical Formulations

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To account for lead time variability (σ_L) and demand variability (σ_d), the dynamic Safety Stock (SS) and Reorder Point (ROP) are calculated as:

$$SS = Z \times \sqrt{\bar{L} \cdot \sigma_d^2 + \bar{d}^2 \cdot \sigma_L^2} \tag{1}$$

$$ROP = (\bar{d} \times \bar{L}) + SS \tag{2}$$

Where $\bar{d}$ represents average daily demand, $\bar{L}$ is the predicted mean lead time, and Z corresponds to the target service level factor.

3 Dataset

The dataset that will be used in this project is version 5 of the *DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS* from Mendeley Data. [2] This data set is a comprehensive supply chain data set that covers procurement, production, sales, and commercial distribution. It comes from DataCo Global, a UK-based consultancy company, that provides data and information management consultancy. This specific data set is publicly available for supply chain projects and is well documented, which allows easy implementation.

This paper will use the standard ETL, as opposed to ELT, and follow the basic steps: Extract, Transofrm, and Load. ETL will be easier to load to our database and keep the project manageable by mitigating scale and complexity. It also allows us to cleanly normalize our data, even though we are already using a structured data set, we will still want to do basic transformations and drop any unnecessary columns or rows.

4 Data Extraction

Extraction requires us to pull the data from a non-local source, in this case we will be pulling from kaggle using the kaggle API given that it is well documented and easy to use, while demonstrating rudimentary knowledge of the API calls and non-local extraction.

The main coding language used in this paper will be Python which gives further credence to using the kaggle API, kagglehub. This will help build a modular pipeline that is dataset agnostic and can be used as needed in the future for any kaggle datasets. The API is already designed to work seamlessly in Python NML workflows and it is free to create personal API Tokens. [3]

To create a manageable and more thorough documentation we will use `uv` to add all dependencies and packages for this project.

```
uv add kagglehub

```

Listing 1.1. Adding kagglehub with uv

Once installed, we will load the dataset directly to a Pandas DataFrame. The documentation on kaggle gives a good overview of how to perform this, we will be slightly modifying kaggle's example to handle.

```
import kagglehub
from kagglehub import KaggleDatasetAdapter

# Load a DataFrame with a specific version of a CSV
df = kagglehub.dataset_load(
KaggleDatasetAdapter.PANDAS,
"shashwatwork/dataco-smart-supply-chain-for-big-data-analysis",
"DataCoSupplyChainDataset.csv",
```

```
    )

```

Listing 1.2. Loading Kaggle CSV dataset into a Pandas DataFrame

5 Data Transformation

This will cover all transformations and the rationale behind the transformations - should I inject null values rather than dropping rows with null values, I should have a valid reason for the decisions and document it whether right or wrong

6 Data Loading

This will cover what I loaded my data into, whether I work directly from a .csv file or choose to load it to SQL or some other database. This choice should also be documented and there should be some rationale behind it.

7 ML Model Training

This will cover how I trained the machine learning model, the type of model, how it was trained, the rationale for the model chosen, and whether or not there are limitations to my choice and rationale

8 ML Model Evaluation

This will be the evaluation of my model to find specific limitations so that we can make inferences about when this model may fail, whether it be on edge cases or is even useful for deployment

9 Dashboard Development

To cover an E2E pipeline, we would need a way to interact with the model through some form of GUI - that way, the end-user can interact or at least see the data in a helpful visual format. In essence, the degree is more about analysis and presentation of data rather than my ability to code, so I view this as a vital part. This may also have another subsection about how the dashboard will be deployed

10 Integration And Testing

This will be a rather small section, but because this will likely be done piecemeal and modular, there will need to be some time spent on integration and to give a good overview of how the parts actually connect

11 Web Deployment

I believe a final, perhaps optional step, would be to deploy the project to the web - I would likely lack live-streamed data, but from an end-to-end view covering some form of hosting and deployment will be beneficial from a knowledge point of view and demonstration of skills.

12 Results and Analysis

This is the final part of the body paragraph where I will examine the results of my project and analyze how well it performs; we might simply perform an ad hoc evaluation or give it edge cases to see where it fails

13 Conclusion

This is the conclusion, broadly summarizing the work that was completed, the results, and analysis. This section should also at least have suggestions for more room for improvement, as that segue's directly into our limitations.

14 Limitations

While I will likely give limitations throughout the paper, having a limitations section at the end of the paper can summarize the found limitations and any other proposed larger limitations

[1]

References

1. Bandi, A.: Operational data pipelines in modern supply chains. Journal of Data Analytics **12**(2), 45–58 (2026)
2. Constante, F., Silva, F., Pereira, A.: DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS (2019). https://doi.org/10.17632/8gx2fvg2k6.5
3. Kaggle: How to Use Kaggle. https://github.com/Kaggle/kagglehub/blob/main/README.md (2026)